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|  <b>Marwadi</b><br>University | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                  |
| <b>Digital Signal and Image Processing (01CT1513)</b>                                                          | <b>Aim : Audio Convolution</b>                                                                                             |                                  |
| <b>Open Ended Problem : 1</b>                                                                                  | <b>Date: 18/07/2026</b>                                                                                                    | <b>Enrolment No: 92400133125</b> |

### **AIM : Audio Concolution**

#### **Code :**

```

from google.colab import files

from pydub import AudioSegment

import numpy as np

import matplotlib.pyplot as plt

from scipy.signal import convolve

# Upload MP3 file

uploaded = files.upload()

# Get uploaded filename

filename = list(uploaded.keys())[0]

# Load MP3 file

audio = AudioSegment.from_mp3(filename)

# Convert to mono and extract raw samples

audio = audio.set_channels(1)

samples = np.array(audio.get_array_of_samples(), dtype=np.float32)

# Define convolution kernel

kernel = np.array([1, 0, 1, 0, 1], dtype=np.float32)

# Perform convolution

convoluted = convolve(samples, kernel, mode='same')

# Normalize

convoluted = convoluted / np.max(np.abs(convoluted))

convoluted_int16 = (convoluted * 32767).astype(np.int16)

```

|                                                                                                                |                                                                                                                            |                                  |
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# Save as WAV

```
convoluted_audio = AudioSegment(
```

```
    convoluted_int16.tobytes(),
```

```
    frame_rate=audio.frame_rate,
```

```
    sample_width=2,
```

```
    channels=1
```

```
)
```

# Export output

```
convoluted_audio.export("output_convoluted.wav", format="wav")
```

```
print("Convolved audio saved as 'output_convoluted.wav'.")
```

# Plot original and convoluted signals

```
samples_norm = samples / np.max(np.abs(samples))
```

```
plt.figure(figsize=(12, 6))
```

```
plt.subplot(2, 1, 1)
```

```
plt.plot(samples_norm, color='blue')
```

```
plt.title("Original Audio Signal")
```

```
plt.subplot(2, 1, 2)
```

```
plt.plot(convoluted, color='red')
```

```
plt.title("Convolved Audio Signal with Kernel [1, 0, 1, 0, 1]")
```

```
plt.tight_layout()
```

```
plt.show()
```

|                                                                                                             |                                                                                                                            |                                  |
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**Output :**

